Adding, Subtracting and Multiplying Polynomials
Answer Key

Algebra 1

Quick Answers

1. $5x^2 - 3x$
2. $5x + 5$
3. $12x^3 - 21x^2$
4. $x^3 + 4x^2 - 5$
5. $x^2 + 6x - 27$
6. $4x^2 - 5x + 11$
7. $6x^2 - 7x - 20$
8. $3x^3 + 6x^2 - 9$
9. $4x^2 + 8x - 5$
10. $x^3 + x^2 - 10x + 8$
11. $3x^2 + 12x + 10$
12. (a) perimeter $= 6x + 16$, (b) area $= 2x^2 + 11x + 15$, (c) = —

What is verified

13 of 14 answers machine-checked against SymPy, across 12 problems.
— marks an answer only you can judge — problem 12. The worked solution states what a correct response must contain.

Curriculum

Level: Algebra 1
HSA-APR.A.1 – *problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12*
Difficulty 1–5 of 5 across 14 tagged checks.

1. Simplify $(3x^2 + 5x) + (2x^2 - 8x)$.

Solution: Adding, so the brackets just come off. Collect the x^2 terms ($3x^2+2x^2 = 5x^2$) and the x terms ($5x - 8x = -3x$). $\boxed{5x^2 - 3x}$

2. Simplify $(7x - 4) - (2x - 9)$.

Solution: Subtracting: change every sign inside the second bracket first, so $-(2x-9)$ becomes $-2x + 9$. Then $7x - 2x = 5x$ and $-4 + 9 = 5$. The -9 turning into $+9$ is the step students skip. $\boxed{5x + 5}$

3. Multiply $3x^2(4x - 7)$.

Solution: One term outside, two inside, so two products. $3x^2 \cdot 4x = 12x^3$ (multiply the numbers, add the exponents) and $3x^2 \cdot (-7) = -21x^2$. $\boxed{12x^3 - 21x^2}$

---

4. Simplify $(x^3 - 2x + 6) + (4x^2 + 2x - 11)$.
Solution: Drop the brackets and line up like terms. There is only one cubic term, x^3, and only one quadratic term, $4x^2$. The linear terms cancel: $-2x + 2x = 0$. The constants give $6 - 11 = -5$. $\boxed{x^3 + 4x^2 - 5}$

---

5. Multiply $(x + 9)(x - 3)$.
Solution:

$$\begin{aligned}(x+9)(x-3) &= x \cdot x + x \cdot (-5) + 9 \cdot x + 9 \cdot (-3) \\ &= x^2 - 3x + 9x - 27 \;=\; x^2 + 6x - 27.\end{aligned}$$

The middle coefficient is $9 + (-3) = 6$ and the constant is $9 \cdot (-3) = -27$. $\boxed{x^2 + 6x - 27}$

---

6. Simplify $(5x^2 - x + 3) - (x^2 + 4x - 8)$.
Solution: Distribute the minus sign across all three terms: $-(x^2 + 4x - 8) = -x^2 - 4x + 8$. Then $5x^2 - x^2 = 4x^2$, $-x - 4x = -5x$, $3 + 8 = 11$. $\boxed{4x^2 - 5x + 11}$

---

7. Multiply $(2x - 5)(3x + 4)$.
Solution:

$$\begin{aligned}(2x-5)(3x+4) &= 6x^2 + 8x - 15x - 20 \\ &= 6x^2 - 7x - 20.\end{aligned}$$

Watch the middle: $+8x$ and $-15x$ combine to $-7x$, not $-23x$. $\boxed{6x^2 - 7x - 20}$

---

8. Add $(4x^3 - x) + (x^3 + 6x^2) + (-2x^3 + x - 9)$.
Solution: Three brackets, all added, so all the brackets come off. Cubes: $4x^3 + x^3 - 2x^3 = 3x^3$. Squares: only $6x^2$. Linear: $-x + x = 0$. Constant: -9. Missing powers simply contribute nothing. $\boxed{3x^3 + 6x^2 - 9}$

---

9. Subtract $(2x^2 - 7x + 1)$ from $(6x^2 + x - 4)$.
Solution: "Subtract A from B" means $B - A$, so the expression is $(6x^2 + x - 4) - (2x^2 - 7x + 1)$ — starting from the wrong one negates every part of the answer. Distributing the minus gives $-2x^2 + 7x - 1$, and then $6x^2 - 2x^2 = 4x^2$, $x + 7x = 8x$, $-4 - 1 = -5$.
$\boxed{4x^2 + 8x - 5}$

---

10. Multiply $(x + 4)(x^2 - 3x + 2)$.

Solution:

$$x(x^2 - 3x + 2) = x^3 - 3x^2 + 2x,$$
$$4(x^2 - 3x + 2) = 4x^2 - 12x + 8,$$

and adding those two rows:

$$x^3 + (-3x^2 + 4x^2) + (2x - 12x) + 8 = x^3 + x^2 - 10x + 8.$$

$$\boxed{x^3 + x^2 - 10x + 8}$$

---

11. Simplify $(2x + 3)^2 - (x - 1)(x + 1)$.
Solution:

$$(2x + 3)^2 = 4x^2 + 12x + 9,$$

not $4x^2 + 9$: the middle term is $2 \cdot 2x \cdot 3 = 12x$. Next,

$$(x - 1)(x + 1) = x^2 - 1,$$

and subtracting the whole second product:

$$(4x^2 + 12x + 9) - (x^2 - 1) = 4x^2 + 12x + 9 - x^2 + 1 \;=\; 3x^2 + 12x + 10.$$

$$\boxed{3x^2 + 12x + 10}$$

---

12. Rectangle with length $(2x+5)$ and width $(x+3)$: perimeter, area, and why the degrees differ.
Solution (a): Perimeter is two lengths plus two widths: $2(2x + 5) + 2(x + 3) = 4x + 10 + 2x + 6 = 6x + 16$. $\boxed{6x + 16}$
Solution (b): Area is length times width: $(2x + 5)(x + 3) = 2x^2 + 6x + 5x + 15 = 2x^2 + 11x + 15$. $\boxed{2x^2 + 11x + 15}$
Solution (c) — instructor-judged. Adding polynomials only ever collects like terms, and collecting like terms cannot create a power that was not already there, so a sum of degree-1 expressions stays degree 1. Multiplying pairs the leading terms with each other: $2x \cdot x = 2x^2$, a power that neither factor had on its own, so the degrees add, $1 + 1 = 2$. Full credit contrasts the two operations; naming only the numbers earns half credit.